13/01/26

Tutorial - 2

Q. A mobile service provider wants to classify sms message as spam or not spam using a simple neural network (Perceptron)

x_1 = specific words and phrases

x_2 = Links and Attachments

x_3 = Bias

x_1	x_2	x_3	Targets (y)
4	2	1	1
1	0	1	0
6	3	1	1
2	1	1	0

Solution :-

Perceptron (ML model)

1) Inputs : x_1 & x_2

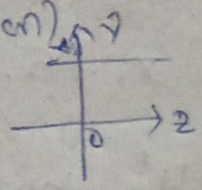
2) weights : w_1 & w_2

3) Bias (b)

4) weighted sum (z) $z = x_1 w_1 + x_2 w_2 + b = \sum_{i=1}^{2} x_i w_i + b$

5) Activation function :- (Here: step function)

$$\hat{y} = \begin{cases} 0 ; & z < 0 \\ 1 ; & z \geq 0 \end{cases}$$



6) updated weights

$$\boxed{w_i^{new} = w_i^{old} + \eta (y - \hat{y}) x_i}$$

η = learning rate = 1

y = Actual or True Target

$\hat{y}$ = Predicted Target

(i) $x_1 = 4$, $x_2 = 2$, $b = 1$

let $w_1 = 0$, $w_2 = 0$

$z = w_1 x_1 + w_2 x_2 + b = 1$ $\hat{y} = 1$

Prediction is correct

Iteration = Epoch

Q2) An email service provider wants to classify incoming emails as spam or not spam using a single network model the training dataset is shown in below

x_1	x_2	x_3	Target (y)
5	3	1	0
1	0	1	0
7	5	1	1
2	2	1	0
6	1	1	1

(i) $x_1 = 5$, $x_2 = 3$, $b = 1$

let $w_1 = 0$, $w_2 = 0$

$z = 1$ $\hat{y} = 1$ $y = 0$ $\eta = 1$

$w_1^{new} = 0 + 1(0 - 1) \times 5 = -5$

$w_2^{new} = 0 + 1(0 - 1) \times 3 = -3$

now,

$w_1 = -5, \quad w_2 = -3 \quad x_1 = 1, \quad x_2 = 0, \quad b = 1$

$z = -5 + 1 = -4 \quad \hat{y} = 0 \quad y = 0$

prediction is correct

Q3) An e-commerce company predicts whether a customer will make a purchase base on time spent on website (min) and no of items viewed

Time on site (x_1)	Items viewed (x_2)	Bias (x_3)	Target (y)
2	3	1	0
15	25	1	1
8	12	1	1
3	5	1	0
12	20	1	1

Sol:-

(i) $x_1 = 2, \quad x_2 = 3, \quad b = 1, \quad w_1 = 0, \quad w_2 = 0$

$z = 2 \times 0 + 3 \times 0 + 1$

$= 1 \quad \hat{y} = 1, \quad y = 0 \quad \eta = 1$

$w_1^{new} = 0 + 1\,[0 - 1] \times 2 = -2$

$w_2^{new} = 0 + 1\,(0 - 1) \times 3 = -3$

Now,

$w_1 = -2$, $w_2 = -3$, $x_1 = 15$, $x_2 = 25$, $b = 1$

$z = -2 \times 15 + -3 \times 25 + 1 = -104$

$\hat{y} = 0$, $y = 1$

$w_1^{new} = -2 + 1[1-0] \times 15 = 13$

$w_2^{new} = -8 + 1(1-0) \times 25 = 22$

Now

$w_1 = 13$, $w_2 = 22$, $x_1 = 8$, $x_2 = 12$, $b = 1$

$z = 13 \times 8 + 22 \times 12 + 1 = 369$

$\hat{y} = 1$, $y = 1$

Prediction is correct